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System Requirements

The following table summarizes the operating system requirements and formats provided for Circa. Circa is a 64-bit plugin.

Operating System	Minimum Version	Formats
macOS	10.13	AAX, AudioUnit, CLAP, VST3; Intel and Apple Silicon
Windows	10.0 version 1607	AAX, CLAP, VST3
Ubuntu	20.0	CLAP, LV2, VST3
iOS (separate purchase)	iOS 16	AUv3

Demonstration Version

We encourage you to download and try the demonstration version of Circa before purchasing it. The demo version of Circa is the same as the regular version, but has the following limitations:

- Presets cannot be saved, nor can parameter values or other settings. This includes the information usually stored by your host DAW. If you save a DAW session with an instance of the demo version of Circa, the plugin will revert to its default state when you reload the session.
- Circa will cease to emit audio altogether 20 minutes after you add it to your DAW session. You can remove it and add it again, but it will revert to its default state.

If you purchase Circa after using the demonstration version, simply run the installer provided to you after your purchase to replace the demo version with the full version.

Introduction

Thank you for purchasing Circa, our creative, multi-channel looping instrument, carefully designed to encourage experimentation and improvisation. With six independent looping layers, one-button operation, sophisticated MIDI mapping, and built-in effects, Circa combines precision and playfulness to inspire your music-making. Whether you're capturing spontaneous performances, crafting complex rhythmic patterns, or creating evolving textures, Circa offers a musical, hands-on workflow with sophisticated capabilities at your fingertips. This manual guides you through Circa's features and possibilities, from basic recording techniques to detailed parameter descriptions, helping you quickly master the plugin and unlock its full creative potential.

Circa Versus Enso

If you've been following our activities (thank you!), you may be wondering about the relationship between Circa and our other loop-oriented plugin, Enso. First, Circa is not a replacement for Enso. Enso's development will continue along its own path, as will Circa's. Enso was originally conceived to replicate a specific tape-based delay system used by Brian Eno and Robert Fripp, starting in the 1970s with the album *Discreet Music*. Our inspiration for Circa, on the other hand, came from contemporary, digital looping devices, ranging from simple guitar pedals to BOSS's sophisticated RC-505mkII Loop Station¹ and various loop-based instruments shared within the generous community of Monome Norns users.

Perhaps the most significant distinction is that Enso employs a single buffer in which it records, plays, and overdubs audio. Circa, on the other hand, has six such buffers which operate independently or in synchrony. Enso has more tools for varying the speed at which it records and plays and possesses the unusual ability to record and play at different speeds simultaneously. Circa's transport controls for individual layers are simpler but provide the ability to automatically adjust recording durations to keep the layers in sync. Enso has tape-like effects for altering the audio it contains. Circa's effects are more mix-oriented, applying to some or all of the layers as desired.

Both tools have their uses, and we would be delighted if you'd take advantage of our demo versions to try both. If you're an Enso user who's used multiple instances of Enso and wished for a smoother way to use them together, Circa may be what you've been wishing for.

Nomenclature

Circa is equally at home on desktop computers, laptops, and tablets. We kept touch-oriented operation firmly in mind while designing and developing Circa. By convention, "clicking" means making an onscreen gesture by pressing the button on a

¹ Product name is a trademark of Roland Corporation.

mechanical device such as a mouse, while “tapping” means making a similar gesture with your finger. For brevity we’re prone to simply saying clicking in this manual, but you can always interpret us to mean tapping as well.

What’s New in Version 1.2

Release 1.2 is the largest update to Circa since it first shipped. It is mostly a matter of the looping engine doing what it always claimed to do: cleaner varispeed, loop seams that actually crossfade, undo you can trust, and a long list of workflow corrections. If you have been living with a rough edge since 1.0, there is a good chance it has been filed off here. There are new capabilities too: the modulation slots can now hold LFOs, and there is a new way to perform with slices.

If you read only one part of this section, read *Changes in behavior* at the end. A few things now work differently on purpose.

Sound quality

- Varispeed is dramatically cleaner. Playing a layer at raised speeds no longer has the harsh, inharmonic grit that bright material used to acquire, and lowered speeds are smoothly filtered rather than gritty. The same improvement applies to *recording* at speeds other than +1, and sweeping the speed through normal no longer produces the fine stair-step crackle it used to.
- Loop seams crossfade properly. Circa records a short hidden tail past the end of every loop and uses it to smooth the splice. A long-standing bug left that crossfade silently inactive, so loops ending in a transient, or in sustained high frequencies, never got the smoothing they were designed to get. It works now: sustained material loops seamlessly with Record In and Out times in the region of 20 to 100 milliseconds.
- Undo no longer dips in volume. Undoing an overdub while playing used to sag, or even drop out entirely for a moment, partway through the transition. The crossfade now holds constant loudness from beginning to end, and the handover at the end is seamless.
- The reverb no longer freezes. Turning the reverb’s output to zero used to trap the tail, so turning it back up seconds later resumed a stale one. The reverb now empties at zero and returns silent until new signal arrives. The Color and Diffusion knobs also adjust smoothly instead of stepping.
- Recording from the internal buss is properly stereo. A layer with its input set to Buss used to record the left-channel sum on both channels, and the right-hand buss meter lied about it. Panning now survives the trip through the buss.
- The output crossfader fully isolates the A and B busses. Signal assigned to A no longer bleeds through when the crossfader is pushed all the way to B.
- Toggling the EQ on and off no longer changes the high-shelf frequency selection, and a few subtler EQ and compressor problems are fixed: the dry/wet mix could blend slightly wrong, and bypassing after a sample-rate change could fade incorrectly.
- The click at the start of playback with Env To Speed engaged is gone.
- Reversed loops no longer click at the loop point. Playing a layer backwards had always spliced rather than crossfaded there (the smoothing only ever applied going forwards) and the join put the loop’s faded-in start against its full-level end, overshooting by half again. Reverse now gets the same crossfade forward playback does, at any speed and with the markers moved in.
- The region between the start and end markers is crossfaded too, which it never was before. Moving the markers inward used to leave a plain splice that clicked once per loop, and dragging a marker through the play head threw the head to the far end of the region; it now shoves the head cleanly along ahead of it. This covers overdubbing layers as well: with the Post-Record switch set to Overdub, a take dropped straight into overdubbing and got no seam crossfade at all, so moving the markers clicked at every pass, and pressing Play afterwards did not put it right.
- Loop seams are clean at slow speeds, including while you drag the markers, and short marked regions keep their crossfades rather than having them squeezed away.
- Layers keep their bottom octave. The DC-blocking filter on every layer’s output sat an octave and a half too high, and got worse at higher sampling rates. It is now a true 10Hz DC blocker at every sample rate, so bass-heavy loops sound the way you recorded them. A fix for a related problem makes moving the play position by clicking in the waveform click-free for speeds less than 1.
- Speed changes no longer spit high frequencies. Playing above +1 correctly filters material that would otherwise alias, but that filter used to swing wide open while the speed was on its way from one value to another, briefly passing the very content it was there to remove. It was most obvious with Damping turned up, where the sweep is slower. Ramps and reversals are clean now.
- Changing a layer’s speed no longer clicks. Reversing direction, or jumping the speed between values, stepped the playback position by a few samples and ticked every time, most obviously when a footswitch or controller flips the direction.

- Clicking in the waveform no longer swoops in pitch when Damping is set to zero.

Recording, looping, and undo

- Long recordings are stable. We fixed random crashes that could occur during takes longer than about six seconds while the waveform display was animating. We believe this explains the rare and previously unexplained crashes some of you saw during long takes. Dropouts around the six-second mark while other layers play are also gone, as are dropouts in the seconds after undoing a long loop.
- Undo is trustworthy now. Undoing an overdub restores the pre-overdub material exactly, every time, including the case of briefly re-entered overdub before undoing which previously left the overdub audible even though the waveform looked right. Rapid undo, redo, and overdub cycles are safe.
- Undo after loading a session no longer erases your loop. Previously, loading a session or preset containing audio (or dragging in a sample), overdubbing, and pressing Undo cut the layer to permanent silence.
- Sample-rate changes no longer destroy loops. Raising your audio device's sample rate mid-session, say from 48 to 88.2 kHz, could silently truncate the back portion of your loops, longer ones especially. Loops now survive rate changes intact in both directions.
- Erase no longer takes material with it when the play head jumps. Holding Erase while a slice jumps, an RTZ, a gesture or a click in the waveform moved the play head erased everything between the two positions, instantly. Erase now simply carries on at the new position, and no longer reaches past the end marker into material outside the played region.
- Dropping a sample onto a layer that is currently playing is safe. It used to be a good way to crash the plugin. Expect a brief silence while the file loads.

Saving and exporting

- You can save while recording. Saving a preset or session during a layer's very first recording now captures the take so far: the file stores everything recorded up to the moment of the save, as a finished, stopped loop with a correctly derived tempo—and the take itself keeps recording, uninterrupted. Previously a save during a first take stored no audio at all for that layer, and saving during any recording risked a corrupted file or a crash.
- Exporting respects your markers. Dragging a loop out of Circa now exports exactly the region between the start and end markers. Previously the exported file contained the entire loop no matter where you had put the markers. With the markers at their default positions, you get the whole loop as before.
- Repeated saves no longer erode loops. Loops loaded from an audio file, and recordings stopped directly from record without passing through play or overdub, used to get imperceptibly shorter—by a single sample—on every save and reload, slowly drifting the loop point over many cycles.

Env to Speed and scrubbing

The tape-style speed envelope got a full overhaul. If you use Env To Speed, particularly with long envelope times, this is the section to try out.

- The playhead tracks the tape. During spin-up and spin-down the position indicator now crawls along with what you actually hear. Previously every speed ramp left the display—and everything driven by it—running ahead of the audio by up to half the envelope time, permanently.
- Overdubs land where you hear them, not at the formerly offset display position.
- Undo is position-continuous: it no longer teleports playback to the start of the loop, but continues from where the transition ended.
- Jumping between slices no longer silences the layer for the duration of the jump's crossfade, and switching Env To Speed on or off while a layer plays now takes effect immediately rather than waiting for you to stop and restart the layer.
- Scrubbing a playing layer now actually ducks, and releasing it is clean. The level drop was only ever applied for the first 10 milliseconds of a scrub, so a held scrub played at full level and letting go produced a momentary 10dB dip. The layer now stays ducked for as long as you are scrubbing and comes back up smoothly.
- Scrubbing had a bug that could permanently silence a layer: the first scrub while stopped, after recording a layer's first take, did nothing, and the layer then stayed silent until you loaded a preset. Fixed. The first stopped scrub now sweeps from the current position rather than sweeping in from the end of the loop, and scrubbing works in Env To Speed mode, where previously only the playhead moved.

Synchronization

- Layers following Master RTZ restart cleanly when the master's recording ends. They now keep sounding through the master's record fade and restart from their beginning exactly once, at the moment the master's loop closes. Previously they fell silent for the whole record fade—barely noticeable with short fades, but up to a second of dropout with long ones.
- Host-synced layers return to zero on every bar-line resume, including when the host starts from the very beginning of the timeline.
- Stopping your DAW's transport now stops Circa's layers. With Host Sync on, a playing layer used to carry on by itself after the host stopped.
- A layer waiting for the next barline pulses its Trigger or Stop button so you can see that it is armed and that your click wasn't ignored: yellow while it waits to record, and the play and stop colors while a play or a stop waits its turn.
- The playhead behaves during the record-stop fade, tracking playback from the top of the loop instead of sitting at the loop's end until the fade completes.

Gestures

- Gesture recording is isolated. Playing one gesture while recording another no longer copies the playing gesture's moves into the new recording, and a gesture can no longer re-record its own playback. What you actually touch is what gets captured.
- Gesture playback no longer floods your DAW. Replayed knob moves used to reach the host up to thirty times a second per parameter, filling its undo history with fictitious edits and writing unwanted automation when the track was record-armed. Playback now moves Circa's controls with virtually nothing reaching the host. Knobs you drag by hand still automate and undo normally.
- Your DAW stays in sync anyway: when a gesture stops, the host is brought up to date with one clean edit per parameter the gesture touched, so automation lanes and generic views show the true final values. If you happen to be dragging one of those controls at that instant, your hand wins.
- The trim (scissors) button on a recording or playing gesture now plays the gesture rather than stopping it.

Control and modulation

- New: the modulation slots can be LFOs. Any of the thirty connections on the Modulations page can now be set to LFO instead of MIDI. Each LFO has its own settings: rate, shape, free-running or synced, phase, and amplitude. The destinations are a subset of the continuous targets for MIDI modulators, tailored to suit the continuous nature of LFO signals. An LFO adds its output to the control it's aimed at rather than taking the control over, so your setting stays where you put it and becomes the center of the swing. Two LFOs aimed at the same destination add together. Each has its own on/off switch, so you can hear what one is contributing without dismantling it. The panel draws the shape moving at its real rate, and so does the closed rectangle, so you can watch all of them at once. The Modulations section of this manual describes them in full.
- New: Hold Slice. A momentary function meant for a footswitch: while it is held, the layer loops just the slice the play head was in, and releasing it carries on from there. Assign it in a single connection with Hold Slice On as the Control Down command and Hold Slice Off as the Control Up command. Use Hold Slice Toggle with switches that don't latch. It works forwards, in reverse, and with Env To Speed engaged.
- Slice number is a new continuous destination, for a controller or for an LFO, and slice jumps now follow your markers: Move To Slice # and Play Slice # count the slices you can see in the waveform display rather than dividing the whole loop, so with the markers moved in they land where you would expect.
- Slice-count shortcuts: the four small numbers above a layer's Slices knob set the slice count to useful values with one tap, saving you the fiddle of landing on them with the knob.
- The controller value at which a momentary MIDI assignment counts as pressed is now yours to set, in the Settings view. It still defaults to 64.
- Selecting a layer from a monome grid, or from a replayed gesture, no longer lands one layer behind.
- New modulation and MIDI-assignment targets. Momentary: mute and solo, on, off and toggle, per layer. Continuous: each layer's delay and reverb send levels and A/B buss selector; the delay and reverb output levels and A/B selectors; and each layer's filter frequency, resonance and type. Frequency and resonance sweep smoothly under a controller, and switching the filter type mid-sweep changes the response instantly, just as the TYPE selector does.
- Transport commands sent from MIDI mappings now do exactly what they say. Play, stop, record and overdub mappings were erroneously behaving as toggles.

- Hold-style MIDI mappings fire once. Activating a control-hold mapping no longer triggers a second action when you release the control, except for a command that releases a held state, Erase Off or Hold Slice Off, which is always delivered, so a connection carrying one of those can't strand a layer erasing or looping a single slice.
- Stop always wins. A Stop command is no longer ignored when a Play command is pending.
- The popup menus grow columns rather than running off the bottom of the screen, which matters on iOS in particular and was becoming necessary anyway given the number of modulation targets.

Presets

- The preset browser gains favorites. Click the star next to a preset's name to mark it, and a Favorites category appears in the leftmost column collecting everything you've marked. The browser's column navigation was overhauled at the same time.

Interface

- The background gradient no longer bands and is drawn once rather than on every repaint, lightening the load Circa's interface puts on your CPU.

Changes in behavior

A few things now work differently on purpose:

- Undo immediately after loading a session, before you have made any overdub, now does essentially nothing instead of erasing the loaded loop. Undo immediately after a *live* first take still erases it, with redo bringing it back, as before.
- Saving mid-recording captures the take so far for the recording layer, as described above. Previously that layer was simply absent from the saved file.
- A gesture that captures an all-layers action, such as All Stop, now replays it as a single all-layers action, exactly as you performed it, rather than as six separate per-layer commands.
- Gesture playback no longer writes host automation. If you were relying on capturing a gesture's output as DAW automation, record the automation from the host side by dragging the controls yourself. The LFOs don't write host automation either, for the same reason.
- With Host Sync on, stopping your DAW's transport stops playing layers, where before it stopped only those that were recording or overdubbing. With Host Sync off, nothing has changed.
- Move To Slice # and Play Slice # count the slices between the start and end markers rather than across the whole loop. If you have mappings that relied on the old behavior with the markers moved in, they will now land somewhere different.

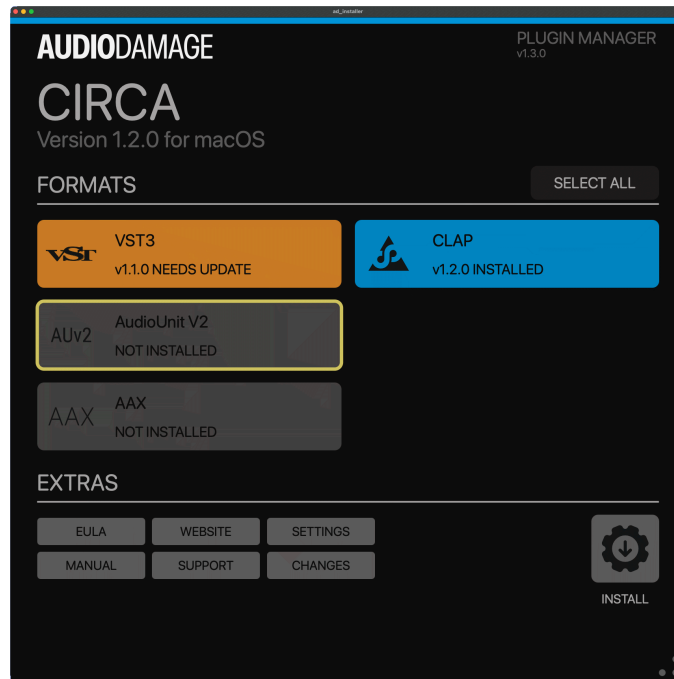
What's New in Version 1.1

If you are coming to 1.2 straight from 1.0, here is what arrived in between:

- The Gesture Recorder records and plays back your manipulations of Circa's controls. Similar to the way Circa records audio, you can record, loop, overdub and play your interactions, augmenting your host's automation features. Gestures can be triggered with MIDI for hands-free recall.
- The play positions of layers became a continuous modulation target.
- Play volume is lowered while scrubbing the position in the waveform display. Yes, it was too loud before. Sorry about that.
- Support for the monome grid and arc controllers, described later in this manual.

Installation

Circa uses our custom plugin manager application for installation. Launch it as usual on your operating system of choice and you'll be presented with a window like this:



Near the top of the window, beneath the name of the plugin, you'll see the version number of the software carried by the installer. This is distinct from the version of the plugin manager itself, which is shown in the upper right and usually not of much interest. (Note that the version numbers in this screenshot are for demonstration purposes only and may not reflect the actual version numbers of the software at the time you read this manual.)

Under the heading FORMATS are large buttons corresponding to the plugin formats which can be installed: AAX, AU, CLAP, LV2 and/or VST3, depending on the operating system. If the plugin is already present on your system in one or more formats (i.e. if you're upgrading from a previous version), the corresponding button is drawn in blue. When possible, the version number of the existing plugin is also shown. If the installer contains a newer version of the plugin than the one(s) already installed, the button is drawn in orange to alert you to the fact that an update is available. If a format is not present on your system, its button is drawn in grey.

Click a button to select the format for installation. A yellow outline appears around the button to indicate that its format will be installed. In the above screenshot, AAX and AudioUnit are not installed, VST3 is installed but older than the version contained in the installer, CLAP is installed and up to date, and the AudioUnit is selected for installation. Clicking a button a second time removes the yellow outline, and the corresponding format will not be installed. Clicking the SELECT ALL button selects all available formats for installation.

No changes to your system's storage device take place until you click the INSTALL button near the lower-right corner of the window. Click that button and you'll receive visual confirmation that the format(s) you've selected have been installed. (Yes, it happens quickly. Our products are not encumbered by any DRM or anti-piracy baggage and hence install more quickly than many others.) On Windows and Linux, if you hold down the Shift key on your keyboard, the INSTALL button's label switches to UNINSTALL, and clicking it will remove the selected formats from your system². Once you've installed and/or removed the formats you need, simply close the application in the usual manner for your operating system. You're done. There is no license code or other authorization necessary; we'd rather assume we can trust you than burden you with an onerous DRM system.

You'll find some handy buttons under the EXTRAS heading, all of which are pretty self-explanatory:

- EULA: presents the End-User License Agreement for our products. By clicking the INSTALL button you're implicitly agreeing to these terms, but we expect that you'll find them reasonable should you take the time to read them.
- MANUAL: opens the current version of this user manual, in PDF form, in your web browser.
- WEBSITE: opens the product's web page in your browser.

²Blame Apple, not us, for the lack of this feature on macOS. On macOS just manually delete the plugin(s) from your plugin folder(s).

- **SUPPORT:** displays information for contacting us, either via our Discord presence or through email.
- **SETTINGS:** produces a panel in which you can choose the installation locations for each plugin format. Note that these locations are nominally stipulated by the publishers of the formats and changing them might cause problems for some hosts.
- **CHANGES:** displays a summary of what we've added, changed or improved since the last version of the product. This button isn't present on the first release of the product.

Operation

Circa appears in your DAW or other host like any other plugin, so we assume that you know the basics of working with plugins in general. Circa processes audio but also receives MIDI. DAWs differ in their handling of routing MIDI to audio plugins; you may have to consult your DAW's documentation for guidance. Bear in mind that you don't have to worry about MIDI routing if you don't intend to control Circa with a hardware MIDI device, and you certainly don't need to worry about it during your initial explorations.

Circa can be used either as a track insert or on an auxiliary channel as a send/return effect. Which configuration you use is up to your personal preferences, but here are a couple of things to consider. If you're using only one sound source with Circa—one software synthesizer, say, or one input on your audio interface—using Circa as a track insert is probably easiest. On the other hand, if you anticipate routing a number of signals to Circa within a given project, or sending more than one signal at once, then putting it on an auxiliary channel and using your DAW's channel-send controls is probably the way to go.

Multiple Outputs

Circa can combine all of its outputs on a single stereo pair or route them to six independent outputs, plus an additional output for monitoring. This lets you process some or all of Circa's layers separately, allowing you to use your choice of effects, EQ, and so on for individual processing. If you haven't worked with a multi-output plugin in your DAW of choice previously you may have to consult the DAW's documentation. Each host seems to have its own idiosyncratic way of handling plugins with multiple outputs, and some are less obvious than others.

Main View

As you've seen by now, Circa looks like this:



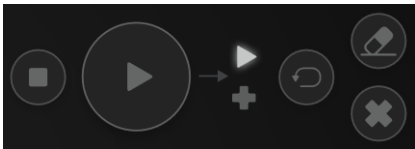
The upper half of Circa's window has four different views, or sets of controls, indicated by the names across the top center: LAYERS, SPEEDS, POSITIONS, and SETTINGS. The lower half of the window has three views: MODULATIONS, EFFECTS, and LEVELS. Click or tap the names to switch between the views. We'll describe each view in turn.

Layers View

Circa's main view is the Layers view, shown above. It's the view that you're likely to use most. Across the top are six circles representing the six layers. Each circle displays a small representation of the audio recorded in the layer, with a radial line showing the current play position within it. Naturally, if no audio has yet been recorded in a layer, its circle is empty.

Clicking or tapping on one of these six circles selects the corresponding layer. The current layer responds to the other controls in the Layers view. In other words, first you choose a layer by clicking on one of the six circles, then you do things with that layer by manipulating the other controls in the Layers view.

To the right of the six circles you'll find controls for recording, erasing, and playing the layer.



From left to right, the controls are:

- **STOP button:** clicking this button stops the layer if it's recording, playing, or overdubbing.
- **TRIGGER button:** this button is Circa's most important control. It does different things depending on what the layer is currently doing. If the layer is empty, click this button to start recording. When the recording is complete, click it again to switch to playing or overdubbing the layer, depending on the state of the POST-RECORD switch described next. If the layer is not empty, click this button to start playing the layer if it's stopped, or to switch between overdubbing and playing. The important thing to remember is that the symbol inside the Trigger button tells you what is going to happen when you click the button. Circa uses the ISO-standard symbols with which you're undoubtedly familiar: a circle for record, a right-pointing triangle for play, and so on.
- **POST-RECORD switch:** this switch chooses what happens when you click the Trigger button the second time, that is, after you've completed recording in an empty layer. It has two settings, **PLAY** and **OVERDUB**; the brighter symbol indicates the active setting. Click either of the symbols to toggle back and forth between the two.
- **UNDO button:** click this button to undo the effect of the most recent recording or overdubbing. If you click it after the first recording, the audio is erased but the length of the recording is preserved. If you click it while overdubbing, the audio you added since clicking the Trigger button—that is, since you switched from playing to overdubbing—is removed. Clicking the Undo button again reverses whatever happened in response to your first click. In other words, it's an undo/redo button.
- **ERASE button:** clicking and holding the upper button on the far right silences the audio traversed by the play head. Use it to selectively remove parts of the layer's recording.
- **DELETE button:** clicking the button with the X symbol erases the entire layer's audio but leaves the length of the layer intact.

The Trigger button, and the audio display below it which we'll get to shortly, change color to reflect the layer's current operating mode or state, as follows:

- Grey: idle or empty
- White: playing
- Red: recording
- Cyan: overdubbing
- Yellow: waiting to begin recording, used in two of the trigger modes described below

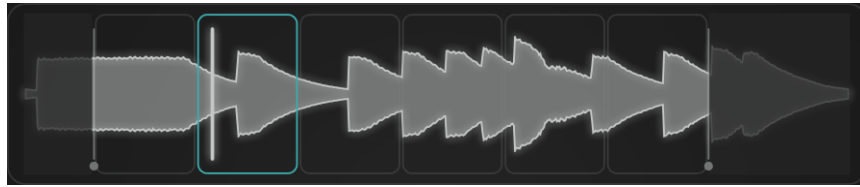
When a layer's Trigger menu is set to Sync, and the layer is waiting for the next barline before an action takes effect, the color pulses gently rather than sitting steady so that you can see the layer is armed and that your click wasn't ignored. The Trigger button pulses yellow while a recording waits its turn and white while a playback waits. The Stop button pulses while waiting for the next barline before stopping.

So, putting together the symbols and the colors, the Trigger button shows you both what is happening now and what will happen after you press it again:

Appearance					
Current state	Idle or empty	Waiting to record	Recording	Playing	Overdubbing
State after click	Recording	Depends on Post-Record	Playing	Overdubbing	Playing

The Waveform Display

The rectangular region beneath the six layer circles shows a graphical depiction of the audio in the layer, or nothing at all if the layer is empty. A vertical line moves across the waveform display as the layer plays, overdubs, or records, indicating the current play/record position. Clicking and dragging within the waveform when the layer is playing moves the play position, allowing you to “scrub” the audio somewhat in the manner of an open-reel tape deck or a turntable.



The small dots at the bottom of the waveform display, and the vertical lines above them, represent the start and end positions of the layer. After the first recording within a layer these positions are set to the far left and right edges of the display, indicating that the entire layer will play over and over again. Click and drag the small dots to move the position markers inward. The left marker determines where the layer will start playing; the right marker sets where playback will jump back to the start marker. In other words, playback, and overdubbing, loop between these position markers.

The SLICES knob to the right of the waveform, when turned up from its default setting of NONE, visually divides the waveform display into equal-sized regions. Rectangles appear on top of the waveform to indicate the slices. When two or more slices are present, clicking or tapping within a slice's rectangle moves the play position to the beginning of the slice. Note that the slices are non-destructive and don't affect the audio in any manner.

You can add a text label to the waveform display to remind you of the layer's purpose or contents; see the Settings View section later in this manual.

Although the waveform display and the layer circles show a single audio waveform, Circa records and plays in stereo.

Speed Control

The wide rectangle at the bottom of the Layers view is Circa's speed control. Clicking and dragging in this rectangle changes the speed at which Circa plays the layer. Circa has a speed range of -2 to $+2$, where $+1$ represents the default speed. Negative values mean that Circa plays backwards relative to the original direction of recording. Like a turntable or tape recorder, changing Circa's speed also changes the perceived pitch of the audio it's playing. So, for example, a speed of $-1/2$ means that Circa plays at half speed, backwards, and an octave below the audio as it was originally recorded, assuming that you recorded it with a speed of $+1$.



On the first recording in an empty layer, Circa implicitly records with positive speeds. It doesn't know in advance how long you're going to record, so it can't start recording at the end of the buffer, since the size of the buffer isn't known in advance. If the speed setting is negative while you're recording, then once the recording is finished Circa will begin playing backwards. Strictly speaking this is something of a contradiction, since Circa should play in the same direction that it moved while recording, but it's more fun if it works as described here, because you hear your loop backwards immediately after recording it.

The switch to the left of the speed control labeled QUANT, as in quantization, causes the speed to jump to one of the values marked on the control: 1 , $-1/2$, and so on. This keeps the relative speeds, and hence the perceived pitches, of Circa's layers in multiples of two so that the layers stay in sync, and in octaves.

Double-clicking the speed control sets the speed to its default value of $+1$.

The knob to the right of the speed control labeled DAMPING imparts a variable amount of drag or sluggishness to the layer's response to speed changes. With zero damping, the speed changes instantly when you click in the speed control. Turning up the damping adds audible pitch slewing when the playback speed changes, reminiscent of the mechanical inertia of turntables or tape players.

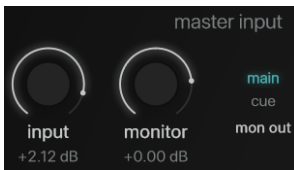
The small triangle at the top of the speed control indicates Circa's current speed. This indicator shows you how much the playback speed lags behind the setting of the speed control with non-zero values of damping.

Drag and Drop

Once you've recorded some audio in a layer, you can copy it to another layer by dragging its circle onto one of the other circles. You can also export a layer from Circa into your host DAW or into a file on your computer's file system by dragging a layer circle onto the appropriate destination. Circa exports exactly the region between the layer's start and end markers, so if you have trimmed a loop with the markers, you get the trimmed version. With the markers in their default positions you get the entire loop. It works the other way, too: dragging an audio clip or file into one of Circa's circles imports the audio into that layer. You can drop a file onto a layer which is currently playing; expect a brief silence while the file loads.

Note: due to restrictions on the AUv3 format imposed by Apple, dragging audio in and out of Circa is not available on iOS. No, there's nothing that we can do about this.

Master Input Controls



The MASTER INPUT section has controls for adjusting the level of signals entering the plugin and for listening to the input signals at the output.

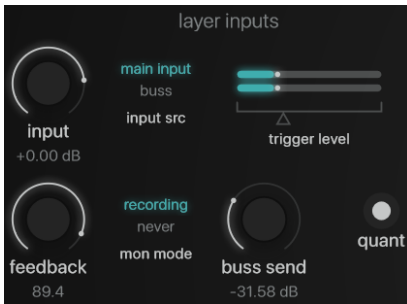
INPUT knob: adjusts the level of the signal as it enters the plugin.

MONITOR knob: adjusts the loudness of the signal passed through the plugin to the output.

MON OUT switch: chooses whether the signal passed through the plugin goes to the main output

buss or the cue buss.

Layer Input Controls



The LAYER INPUTS section contains controls for adjusting and routing the signals arriving at the individual layers for recording. The controls reflect the settings of the current layer, chosen by clicking one of the six circular layer displays at the top of the window.

INPUT knob: adjusts the level of the signal before it is recorded in the layer.

INPUT SRC switch: Circa has an internal signal path for recording signals from one or more layers into another layer. To record signals from this path, set the Input Src switch to BUSS. To record the signals received from the plugin's host, choose MAIN INPUT.

TRIGGER LEVEL slider: the horizontal meters display the level of the signal recorded by the layer. The Trigger Level slider beneath them sets the level at which a layer will start

recording, if its Trigger mode is set to AUDIO. For instance, if there is a bit of noise in your input signal, set the Trigger Level slider above the level of this noise so that the layer won't start recording until you start playing your instrument.

FEEDBACK knob: sets the amount of signal which is retained in the layer during overdubbing. If the Feedback is set below its maximum level of 0 dB, the signal present in the layer will fade out during overdubbing. Putting it at its minimum setting will cause any existing material to be completely erased while overdubbing.

MON MODE switch: chooses whether the signal at the layer's input will be heard when the layer is recording, or not heard at all.

BUSS SEND knob: adds an adjustable amount of the layer's output to the internal signal path. This knob is like an aux send knob on a mixer or in a DAW: the signals from one or more layers are added together with the Buss Send knob, and the summed signal becomes available to any and all layers whose Input Src switch is set to Buss.

You may be wondering about the relationship between the FEEDBACK knob and the BUSS SEND knob, since the latter, in conjunction with the INPUT SRC switch, creates feedback paths within Circa. Despite the nomenclature, they serve very different roles and are not related. The Feedback knob adjusts a signal within the layer: the signal that is played and re-recorded—fed back—when overdubbing new material on the layer. The Buss Send knob, on the other hand, controls a signal sent from the layer to other layers.

Signal Flow

Having just described the input controls in some detail, let's tie the signal-flow room together. Signals entering Circa first encounter the INPUT knob in the Master Input section; turning this knob attenuates or boosts the signal. Often you won't need to touch it, since presumably you have other ways of controlling the level of the signal before it reaches Circa, but this knob can be handy for quick tweaks. Next the signal travels straight to one of Circa's outputs so that you can hear what is going into Circa if you're using it as a track insert. Which output it goes to—either the main output or the cue output—depends on the MON OUT switch. The MONITOR knob controls the loudness of the monitor output.

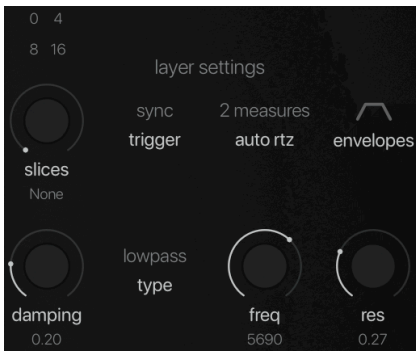
Each of Circa's layers receives either the input signal or a signal from Circa's internal routing buss, depending on the state of the INPUT SRC switch in the Layer Inputs section. This signal can be boosted or cut with the layer's own INPUT knob. The horizontal meters show the relative level of the signal. Bear in mind that, unlike the controls in the Master Input section described previously, the controls in the Layer Inputs section apply to the current layer and can be different for each layer.

The MON MODE switch optionally routes the layer's input signal to the layer's output. This is different from the signal path controlled by the Master Input section, in that the layer's input signal can be passed to its output only when the layer is recording, whereas the master monitoring signal path operates independently of any activities and settings of the layers.

The BUSS SEND knob determines how much, if any, of the layer's output signal is added to the internal routing buss for recording by other layers.

Each layer's overall loudness, stereo position, effects levels, and output assignments are controlled by the mixer-like Levels pane, which we'll describe later.

Layer Settings



The controls in the LAYER SETTINGS section affect how the layer records and plays. There are also controls for the layer's multi-mode filter.

SLICES knob: turning this up from its default value of NONE applies a non-destructive grid to the layer's audio, dividing it into 2 to 16 equal-length slices. Rounded rectangles drawn over the layer's waveform show the slices. Clicking within one of these rectangles moves the play position to the beginning of the slice, whether or not the slice is currently playing. The Slices knob has no effect during recording or overdubbing. The four small numbers above the knob (0, 4, 8 and 16) set the layer's slice count in one tap, 0 meaning None. They're there because those are the counts you reach for most.

TRIGGER menu: sets one of three triggering modes, which work in conjunction with the Trigger button to determine when recording starts in an empty layer. The modes are described below.

AUTO RTZ menu: chooses one of several modes which cause the layer to restart playing from its beginning while playing—in other words, to automatically return to zero. These too are described below.

ENVELOPES: clicking this icon invokes a pane for setting the record and playback envelopes, described below.

ENV TO SPEED switch: causes the Play envelope to affect the playback speed rather than the volume. When you start the layer, the play speed will start at zero and move to the layer's speed setting in the time set by the IN control. Playback slows to zero and stops in the time set by the OUT control after you stop the layer.

DAMPING knob: this knob limits how quickly playback speed changes when you click in the play speed control. Turning the Damping knob up makes speed changes happen more slowly. See the section above on the speed control for more information.

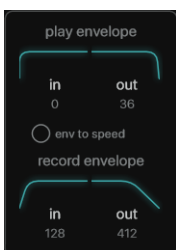
FILTER controls: each layer has its own analog-modeled, multi-mode filter. The TYPE menu chooses the response mode: lowpass, highpass, bandpass, notch, and none, the last of which bypasses the filter altogether. The FREQ knob sets the corner or center frequency of the filter. The RES knob causes the filter to emphasize frequencies near the setting of the Frequency knob. In the case of the notch mode, the Resonance knob narrows the frequency notch, removing less of the signal.

The TRIGGER menu offers three choices:

- **FREE:** recording begins as soon as the Trigger button is pressed.
- **AUDIO:** recording begins when the level of the incoming audio exceeds the setting of the TRIGGER LEVEL slider. In other words, you can hit the Trigger button and recording won't start until you start playing your instrument.
- **SYNC:** recording is synchronized to the playback of the first layer. After you hit the Trigger button, recording starts when the first layer's play position crosses the beginning of the loop. This setting also causes the layer's length to be rounded to large fractions, or multiples, of the length of the first layer. In other words, the Sync option makes it easy to record subsequent layers that are perfectly synced to the first layer.

The AUTO RTZ menu offers these:

- **NEVER:** no effect; the layer plays over and over without interruption.
- **MASTER RTZ:** the layer restarts playing as dictated by the Auto RTZ setting of the first, or master, layer.
- **1, 2, 4 or 8 MEASURES:** the layer restarts playing after the given number of measures, if the global HOST SYNC setting is on and Circa's behavior is determined by your host's transport. If Host Sync is off, a measure is considered to be the length of the first layer.



Clicking the ENVELOPES icon produces a pane for setting the record and playback envelopes. Each has two settings, IN and OUT. Their values are shown in milliseconds; change them by dragging vertically.

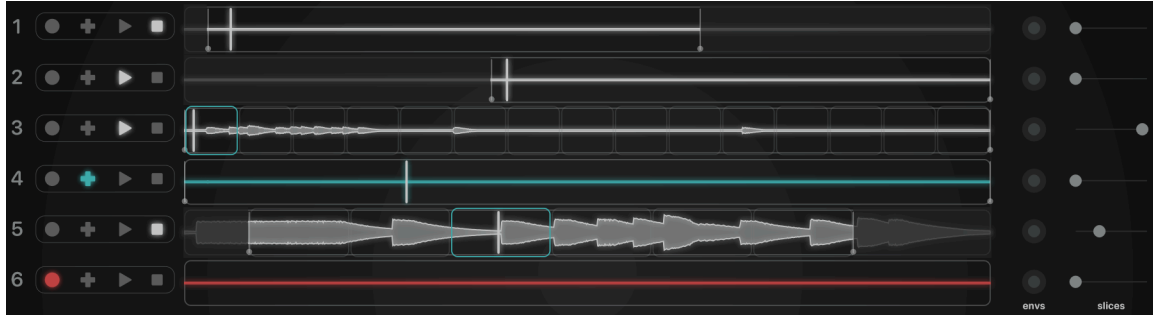
The PLAY envelope creates fade-in and fade-out ramps when playback starts and stops. These volume ramps are also applied when you switch slices by clicking on the layer's waveform. Generally you'll want to use short or zero-length ramps when working with rhythmic or percussive material, to create fast jumps in the loop. Use longer ramps with sustained material for smooth transitions. Jumping in and out of audio with no volume ramps produces clicks; very short ramps help remove these clicks. The Play envelope settings are non-destructive; they do not affect the original recording.

The RECORD envelope, on the other hand, applies volume ramps while the layer is recorded and during overdubbing. When recording or overdubbing starts, the incoming audio fades in over the duration set by the Record In control. When recording stops, the audio fades out for the length of the Record Out control. On the initial recording in an empty layer, this fade-out happens after the end of the loop and is used during playback to crossfade with the beginning of the loop.

Hence you can create seamless loops of sustained material by setting the Record In and Out times to somewhere around 20 to 100 milliseconds. Hold down a chord on your keyboard, hit the Trigger button to begin recording, and continue sustaining the input signal after you hit the Trigger again to finish recording. Circa automatically creates a crossfade at the loop splice using the fade-out it recorded.

Speeds View

The Speeds view shows six instances of the speed control strip found in the Layers view. Each behaves the same as described above, allowing you to manipulate the speed of all six layers rather than just the current layer.



To the left of each control strip you'll find buttons for switching the layers between their various operating states: recording, overdubbing, playing, and stopped. Click or tap the buttons to change states.

On the right of each control strip is a button for turning speed quantization on or off, and a slider for adjusting the speed damping. Both of these controls operate as described previously in the Layers View section.

Positions View

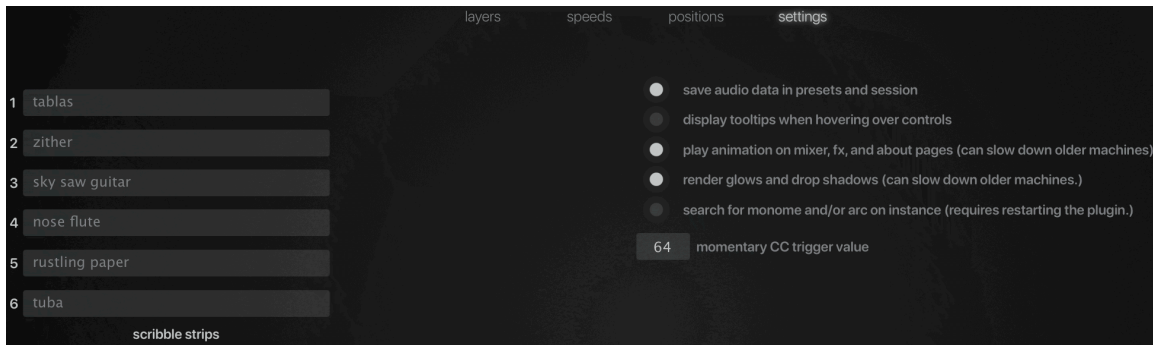
The Positions view contains six instances of the waveform display and attendant controls seen in the Layers view. Each operates in the same manner as the single waveform display in the Layers view, showing you linear depictions of the contents of all six layers in a single view. You can change the start and end markers, move the play position, and scrub in all six displays.



As in the Speeds view, there are buttons on the left for switching the layers between their various operating states: recording, overdubbing, playing, and stopped. Click or tap the buttons to change states.

To the right of the waveform displays are buttons for invoking the Envelope popup panes and a slider for activating slices within the layer.

Settings View



Circa's Settings view contains several controls affecting the behavior of presets and the plugin's appearance.

First, the six text fields on the left, labeled SCRIBBLE STRIPS, allow you to enter names or short comments for each layer. Anything that you type into these fields is superimposed over the waveform displays in the Layers and Positions views.

The switches on the right perform the following functions:

- **SAVE AUDIO DATA IN PRESETS AND SESSION:** if this switch is turned on, the audio recordings in the layers are saved with any presets you save, and in your host DAW's session, along with the plugin's parameters and other settings as usual. If this switch is not on, the audio is discarded when you close your host or the standalone version of Circa. There are a couple of things to consider about this switch. If it's off, the audio is gone for good after you close your DAW. On the other hand, saving the audio means generating potentially large preset and DAW-session files, large enough that you will notice the additional load time when you reopen the session. You may find it preferable to always record the audio emanating from Circa into tracks in your DAW, rather than relying upon Circa to retain the audio. Your DAW is far better suited to storing large chunks of audio than any plugin.
- **DISPLAY TOOLTIPS WHEN HOVERING OVER CONTROLS:** when this switch is on, Circa displays brief descriptions of the controls when you point at them with your mouse cursor. While we hope that you'll find this useful while learning your way around the plugin, we also recognize that you might find it annoying later. Turn this switch off if such annoyance arises.
- **PLAY ANIMATION ON MIXER, FX, AND ABOUT PAGES:** turn this switch on to invoke something nice to look at in the Levels and Effects views and Circa's information pane. Turn it off if it slows down an older machine.
- **RENDER GLOWS AND DROP SHADOWS:** turn this switch off to reduce the complexity of Circa's graphics rendering, which can improve performance on some older computers.
- **SEARCH FOR MONOME AND/OR ARC ON INSTANCE:** enables Circa's integration with monome controller hardware. See the chapter on monome grid and arc control for more information.

Below the switches is a numeric field labeled **MOMENTARY CC TRIGGER VALUE**. It sets the controller value at which a momentary MIDI assignment counts as pressed; any value below it counts as released. The default of 64 suits most controllers, and you can enter anything from 1 to 127. Raise it if a controller of yours sweeps through intermediate values on its way to maximum and trips Circa's commands too early, or lower it if a controller never quite reaches the middle of its range. Like the switches above it, this setting applies to Circa as a whole rather than being stored in individual presets.

Global Controls

There are a few controls present at the top of Circa's window regardless of which view is active.



- **HOST SYNC switch:** if this switch is off, Circa pays no attention to your host DAW's tempo or transport activity. If Circa happens to be your main compositional tool for a project and you don't need tight synchronization between it and other tracks in your DAW, leave this switch off; Circa uses the first layer you record to establish a reference for other layers. On the other hand, if you do want things you record in Circa to stay in sync with other things in your DAW, turn this switch on. See the section on Synchronization and Timing below for more information.
- **ALL STOP button:** click or tap this button to stop any and all layers which are currently playing or recording. A long press on this button resets all of Circa's layers, returning them to their empty, pre-recorded state. Use with caution.
- **ALL PLAY button:** click or tap this button to start playback of all layers which have audio in them.

- **PRESET button:** click or tap this button to invoke the preset browser, described later under Presets. Eventually we'll think of some other symbol for the act of saving information on a computer, but for now we're still resorting to the anachronism of a floppy-disc icon.

Synchronization and Timing

Broadly speaking, Circa can operate in one of two ways: using a host DAW as its timing reference, or using an internal timing reference. As mentioned previously, the HOST SYNC switch near the top of the window chooses one of these two operating modes. If you're using a standalone version of Circa, or if you don't need tight synchronization between Circa's activity and other tracks in your DAW, turn this switch off. On the other hand, if you do want Circa's loops to stay in sync with your DAW, turn this switch on. There are advantages and disadvantages to either setting, which we'll discuss next.

External Sync: Host Sync On

With Host Sync on, Circa relies on information from your DAW for timing. When the layer's TRIGGER menu is set to SYNC, Circa waits for the DAW transport to cross barlines before it responds to the first click on the Trigger button. In other words, you can hit the Trigger button while your DAW plays and Circa won't start recording until your DAW reaches the beginning of the next measure. Once it has started recording, and you hit the Trigger button a second time to finish the recording, Circa will extend or truncate the raw recording so that the layer stays in sync with your DAW when it plays. It sets the length of the recording as follows, with lengths expressed as measures:

Raw Length	Saved Length
3/16 or less	1/8
3/16 to 3/8	1/4
3/8 to 3/4	1/2
more than 3/4	1
more than 1	the nearest whole measure

This looks complex, but what it boils down to is that the recording is rounded to the nearest 1/8, 1/4, 1/2, or full measure, rounding to larger intervals for longer recording lengths. In other words, you don't have to hit the Trigger button with absolute precision; Circa will tighten up the timing of the recording so it plays in sync with the other layers and your other tracks.

Circa remembers the tempo setting of the host at the time you record the first layer. If you change the host's tempo, Circa plays the layers slower or faster to keep them in sync with the rest of your project. It uses a resampling approach for speed changes, so the perceived pitch of your audio goes up and down when played faster or slower. If you need to change the tempo of your project without introducing pitch artifacts, print Circa's output on one or more tracks and use your DAW's more capable tools for time adjustment.

Layers which are synchronized to the host return to zero whenever the host's transport resumes from a bar line, including the very beginning of the timeline.

Stopping your DAW's transport stops Circa's layers along with it, regardless of whether they're playing, recording, or overdubbing. If you'd rather have Circa keep running while you stop and start your DAW, turn Host Sync off.

Internal Sync: Host Sync Off

With Host Sync off, Circa uses the first layer you record as a timing reference. It can be any of the six layers, but it's easiest to keep track of what's going on if you always start with the first, left-most layer. After you record the first layer, Circa makes a guess about how many measures, and at what tempo, this layer represents. It's an educated guess, but it's still just a guess. You may play something that you think of as two measures at 75 beats per minute, which lasts 3.2 seconds. However, one measure at 150 bpm also lasts 3.2 seconds. Circa has no way of knowing which it should be; it just knows that you recorded a chunk of audio that's 3.2 seconds long.

Given this ambiguity, why does Circa guess at a tempo marking and number of measures—why does it even care? Because it uses the same procedure described in the previous section to adjust the lengths of the *second* and subsequent layers you record. The *first* layer serves as the timing reference, so Circa has to decide, for itself, how many measures the first layer represents. After that it can apply the rounding approach described above to extend or shorten the recordings you make on the other layers, assuming that their Trigger menu is set to Sync.

Once the first layer is recorded, Circa uses its measured tempo henceforth. Changing the speed of the first layer does *not* alter the internal tempo setting. This means that if you set the speed of the first layer to something other than +1, the synchronization of

other layers might not behave as you expect. Everything should work well if the first layer's speed is one of the quantized settings, like 1/2, since these are simple multiples of the original speed.

Switching Host Sync On and Off

Switching Host Sync on and off can be done, of course, but there is a complication to be aware of. If you record your first layer with Host Sync off and then turn Host Sync on, you'll almost certainly hear your layers speed up or slow down. This is because Circa adopts the host tempo when you turn sync on—that's what sync is for, after all—and it's unlikely that the tempo that Circa assigned based on the length of your first layer will match the current tempo of your DAW. You can adjust the DAW's tempo to bring Circa's audio back to its original speed and pitch.

But This Is Complicated...

Yes, it is. Most of the time you don't have to think about it much, though. Turn Host Sync on and use the Sync setting in the layer's Trigger menu if you're using Circa in a DAW in conjunction with other instruments and want things to stay lined up. If you're using Circa by itself and just want its layers to play together nicely, again use the Sync setting in the Trigger menu, but turn Host Sync off. Either way Circa will do its best to adjust your recordings so that they play together in a cohesive manner, without requiring you to hit the Trigger button at exactly the right times.

...And Really It's Not That Important To Me

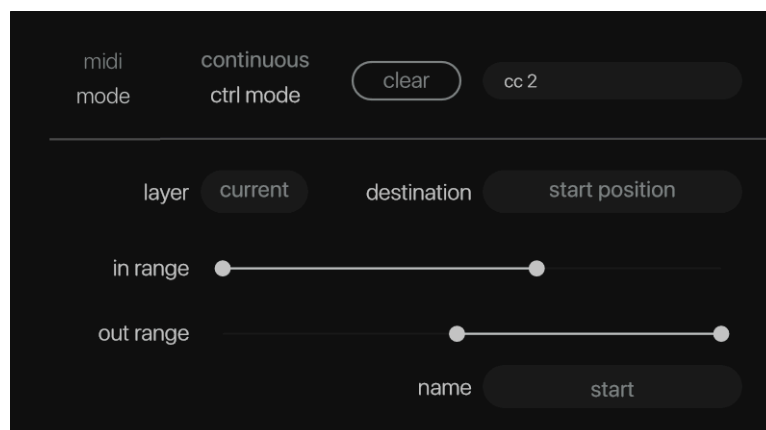
We hear you. One wonderful approach to loop-based composition and improvisation is to not worry about keeping everything in sync, to deliberately record loops of different lengths, and to rejoice in the unexpected juxtapositions that occur as the sounds move together and apart, in phase and out of phase, creating polyrhythms and a whole greater than its parts. If this speaks to you, the other two entries on the Trigger menu, Audio and Free, are for you. With either of these settings, the Host Sync switch is irrelevant. The duration of the recording depends entirely on when you hit the Trigger button; no alteration happens afterwards. Layers recorded this way cycle at their own intervals, independently.

Note that it's not an all-or-nothing sort of affair. You can record some layers with the Sync setting and others with the Audio or Free settings. Sync-enabled layers will keep their rhythmic relationships while the others will drift apart from them and each other—probably, unless your timing is very good. Just remember that the Trigger control applies when the layer is *recorded*; it has no effect after that. Also, remember that the AUTO RTZ control lets you reset any layer, regardless of how it's recorded, at predictable intervals. This can impart rhythmic cohesion on layers originally recorded without synchronization.

Modulations

Circa's modulation system allows control of its looping layers and parameters, either with signals arriving from outside Circa or with signals Circa generates for itself. Each rounded rectangle in the Modulations view represents a connection between a modulation source and a destination. There are thirty slots for connections.

To add a modulation connection, click one of the plus symbols in an empty rectangle. The modulation assignment panel appears.



The MODE menu at the top left of the panel chooses where the connection gets its signal: MIDI, for messages arriving from a controller, or LFO, for one of Circa's own low-frequency oscillators. The rest of the panel changes to suit your choice. MIDI connections are described first and the LFOs afterwards.

To remove a connection altogether, whichever kind it is, click and hold on its rectangle for about one second.

MIDI Connections

A MIDI connection has the following controls:

- **CTRL MODE switch:** this switch has two options, **MOMENTARY** and **CONTINUOUS**. These names reflect the motion of the physical device that you interact with to generate MIDI messages: momentary for things that send discrete messages after a brief interaction, continuous for things that send a stream of messages as you move them. Use Momentary for pushbuttons, switches, keyboard keys, and drum pads. Use Continuous for knobs, sliders, and the like.
- **LEARN button:** after you click this button, Circa waits for the next MIDI message—either a note-on message or a controller message—and records it as the modulation source for this connection. Text indicating whether it's a note message or a controller message, and the note or controller number, appears in the field to the right upon reception of the message, and the button's name changes to **CLEAR**. Clicking it again removes the assignment, should you decide you wanted a different source for this connection.
- **LAYER menu:** a popup menu to choose the layer affected by this connection: the current layer, all layers, or one of the six layers.
- **NAME field:** an editable text field in which you can enter a name to remind yourself of what the modulation does.

The remaining controls depend on the setting of the Ctrl Mode switch. If this switch is set to Momentary, the following controls appear. Each is a popup menu listing the commands sent upon a movement of the assigned controller:

- **CONTROL DOWN:** the command sent upon pressing the assigned key or switch. For controller messages, a value of 64 or greater is interpreted as “down.” (That default threshold value of 64 can be changed with the **MOMENTARY CC TRIGGER VALUE** setting described under Settings View above.) This is usually how buttons and footswitches on MIDI controllers work. If your controller of choice doesn't work this way, you can either change its behavior with software supplied by its maker, or you can assign Circa's commands in an inverted sense.
- **CONTROL UP:** the command sent upon releasing the assigned key or switch. For controller messages, any value below that same threshold is interpreted as “up.”
- **CONTROL HOLD:** the command sent after the assigned key or switch is held down for about a second.

The commands on these popup menus are as follows. Other than the first eight commands, which change which layer is selected, the commands apply to the layer or layers chosen with the Layer popup menu described previously.

- **UNASSIGNED:** nothing happens.
- **INC LAYER:** selects the layer to the right of the current layer. If the rightmost layer was selected upon reception of this command, the leftmost layer will be selected.
- **DEC LAYER:** selects the layer to the left of the current layer. If the leftmost layer was selected upon reception of this command, the rightmost layer will be selected.
- **SEL LAYER 1-6:** makes the layer indicated by the number the current layer, counting from left to right.
- **PLAY:** starts playing, if there's audio in the chosen layers.
- **STOP:** stops playing.
- **PLAY TOGGLE:** starts playing the chosen layers if they're not playing, and stops currently playing layers.
- **RECORD ON:** begins recording in the chosen layers.
- **RECORD OFF:** finishes recording.
- **RECORD TOGGLE:** starts recording the chosen layers if they're not recording, and stops currently recording layers.
- **OVERDUB ON/OFF/TOGGLE:** same as the Record commands above, but turns overdubbing on and off rather than recording.
- **ERASE ON/OFF/TOGGLE:** same as the Record commands above, but turns erasure on and off rather than recording. Note that unlike the eraser button in Circa's window, the Erase On command is not momentary. Erasing will continue indefinitely once it's started. Hence you'll usually want to map Erase Off when you map Erase On. For instance, if you map Erase On to a key press, map Erase Off to the key release in the same assignment.
- **MUTE ON/OFF/TOGGLE:** mutes or unmutes the chosen layers, equivalent to the M buttons in the Levels view.
- **SOLO ON/OFF/TOGGLE:** solos the chosen layers, equivalent to the S buttons in the Levels view.
- **REVERSE DIRECTION:** reverses the play, record, or overdub direction of the chosen layers.
- **ERASE CONTENTS:** destructively silences all of the audio in the chosen layers but leaves their loop length intact.
- **TRIGGER:** equivalent to pressing the Trigger button; initiates recording, playback, or overdubbing.
- **RESET:** returns the chosen layers to their unused state, destructively silencing all of the audio and clearing the loop length.
- **RTZ:** forces the playback position to immediately move to the beginning of the recording of the chosen layers.

- **UNDO:** equivalent to pressing the Undo button; removes the result of the last recording operation in the chosen layers.
- **PLAY GESTURE:** starts playing a recorded gesture. Use the Layer menu to choose which gesture; they're numbered 1 to 6 from left to right just like the layers.
- **STOP GESTURE:** stops a recorded gesture.
- **MOVE TO SLICE #:** moves the play position to one of the layer's slices, leaving the layer otherwise undisturbed. The slices it counts are the ones you can see in the waveform display, between the start and end markers, set with the Slices knob. When the connection's source is a note message, the note you assigned selects the first slice and the notes above it select the slices following it, so a run of adjacent keys or pads falls across the layer's slices in order.
- **PLAY SLICE #:** the same, but also starts the layer playing if it was stopped.
- **HOLD SLICE ON:** the layer loops the single slice the play head was in when the command arrived, repeating it for as long as the hold lasts. Each repetition uses the same crossfade as jumping to a slice by hand. It works forwards, in reverse, and with Env to Speed engaged. With the Slices knob set to None there is only one slice, so there is nothing to hold and nothing happens.
- **HOLD SLICE OFF:** releases the hold. The play head carries on from wherever it had reached within the slice rather than jumping back to where the hold began.
- **HOLD SLICE TOGGLE:** turns the hold on if it's off and off if it's on, for switches that don't latch.

Hold Slice is meant for a footswitch. Assign HOLD SLICE ON as the Control Down command and HOLD SLICE OFF as the Control Up command in the same connection, and the layer stutters on one slice for as long as your foot is down.

If the Ctrl Mode switch is set to Continuous, a single popup menu appears in place of the three above:

- **DESTINATION:** the setting or parameter which changes when this modulation connection receives a control-change message.

The destinations on this menu are as follows. Other than the Layer Select destination, the changes affect the layers chosen with the Layer popup menu or the Layer Select destination. In other words, by using the Layer Select destination, you can use one controller to select the layer which will receive subsequent control-change messages from other controllers.

- **UNASSIGNED:** nothing happens.
- **LAYER SELECT:** changes which layer is selected.
- **SPEED:** sets the speed of the record/playback position.
- **SPEED DAMPING:** same as the DAMPING knob; slows the record/playback position's response to changes.
- **START POSITION:** sets the position from which the chosen layers will start playing, and jump back to from the end position.
- **STOP POSITION:** sets the position at which the chosen layers will stop playing and jump back to the start position.
- **PLAY FADE IN:** same as the In setting of the Play envelope; adjusts the fade-in duration used when playing starts.
- **PLAY FADE OUT:** same as the Out setting of the Play envelope; adjusts the fade-out duration used when playing stops.
- **RECORD FADE IN/OUT:** same as Play Fade In and Out, but affects the Record envelope.
- **INPUT SOURCE:** same as the layer's INPUT SRC switch; chooses whether the layer records from the plugin's input or the internal signal buss.
- **TRIGGERED RECORD LEVEL:** same as the small slider below the input-level meters; adjusts the loudness threshold at which recording will commence.
- **OUTPUT LEVEL:** same as the LEVEL knob in the Levels view; controls the overall loudness of the layer's signal.
- **PANNING:** same as the PAN knob in the Levels view; moves the layer's signal back and forth in the stereo field.
- **BUSS SEND LEVEL:** same as the BUSS SEND knob; adjusts how much of the layer's signal is added to the internal signal buss.
- **POSITION:** moves the play position of the chosen layers. Note that this can set up an argument between the modulation information and Circa's usual control of the position, so the results may or may not be to your liking.
- **CROSSFADER:** the global CROSSFADER in the Levels view, moving the whole mix between the A and B busses. Unlike the other destinations on this list it isn't a per-layer setting, so the Layer menu has no bearing on it.
- **DELAY SEND and REVERB SEND:** same as the send knobs at the top of the layer's channel strip in the Levels view.
- **X-FADE SELECT:** same as the layer's X-FADE selector in the Levels view, choosing whether the layer is assigned to the A side of the crossfader, the B side, or passed through unaffected.
- **DELAY LEVEL and REVERB LEVEL, and DELAY X-FADE and REVERB X-FADE:** the output levels of the two effects and their assignments to the crossfader.

- **FILTER FREQ, FILTER RES, and FILTER TYPE:** the layer's filter controls. Frequency and resonance sweep smoothly under a controller. Switching the filter type mid-sweep changes the response instantly, just as the TYPE selector does.
- **SLICE #:** moves the play position from slice to slice. The controller's travel is divided by the number of slices in the layer, so sweeping it from one end to the other steps the play head through the slices in turn.

Also present when the Ctrl Mode switch is set to Continuous are a pair of range sliders which work together to tailor the modulation connection's mapping of MIDI controller values to the settings they affect:

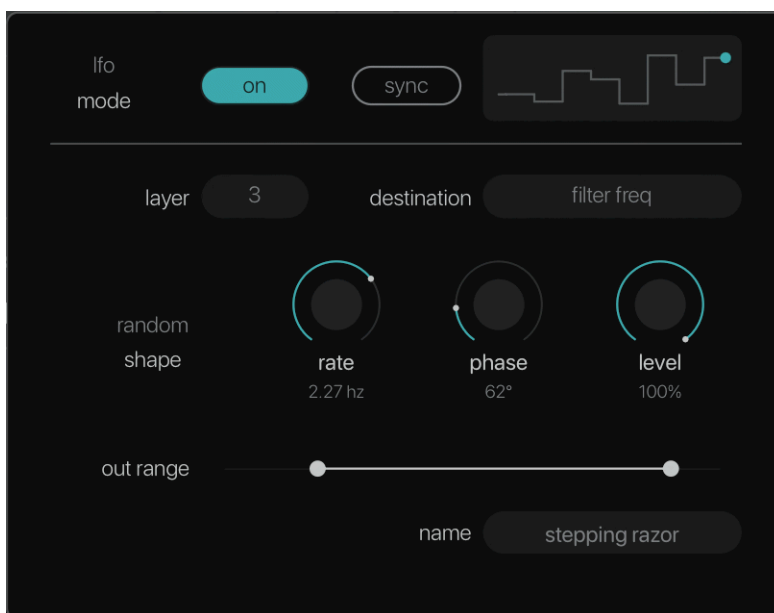
- **IN RANGE:** sets the range of controller values recognized by the connection. The positions of the slider thumbs indicate the lowest and highest values, within the standard range of 0 to 127. Incoming values outside of this range are ignored. For example, if you leave the left thumb at its default position at the far left, and move the right thumb to the middle of the range, only controller values in the range 0 to 64 will cause a change in Circa's settings.
- **OUT RANGE:** confines the values sent to the destination. The slider represents the entire range of the destination's values, and the two thumbs set the lowest and highest values the destination will be driven to. Note that the thumbs clip the outgoing values rather than scaling them: anything that would fall outside the thumbs is held at the nearer of the two. For example, if the In Range thumbs are set as previously described, and the Out Range thumbs are set to positions about 1/3 and 2/3 of the length of the Out Range slider, and the Destination popup menu is set to Start Position, incoming MIDI messages in the range 0 to 64 move the chosen layer's start-position marker within the middle third of the length of the layer's loop, with the lower and upper portions of the incoming range parking it at the one-third and two-thirds points.

The modulation rectangles blink briefly when a MIDI message matching their assignment arrives.

The LFOs

Setting a connection's Mode menu to LFO replaces the MIDI half of the panel with a low-frequency oscillator of Circa's own. There's no learning to do and no controller to plug in; the connection starts running as soon as you've set it up, and drives its destination for as long as it's switched on. Any or all of the thirty modulation connections can be LFOs.

An LFO doesn't take over the control it's pointed at. Circa adds the LFO's output to whatever that control is already set to, so the knob stays where you left it and remains the center of the swing. For instance, move the layer's PAN knob while an LFO sweeps it and the whole sweep moves along with the knob. Point two or more LFOs at the same destination on the same layer and their outputs add together, which is the easiest way to build a motion that doesn't repeat on a predictable cycle.



An LFO connection has the following controls:

- **ON button:** switches the oscillator on and off. Turning it off leaves the connection and all its settings intact but stops it driving anything, and the destination goes back to whatever its own knob says. It's the quickest way to hear what a particular LFO is contributing.
- **SHAPE menu:** the waveform the oscillator produces. TRIANGLE rises and falls in equal measure, SQUARE sits at its high value for the first half of the cycle and its low value for the second, and SINE is the smoothest of the three. SAW falls from high to low across the cycle while RAMP rises from low to high; they're mirror images of one another. RANDOM holds a new value, chosen anywhere in its range, at the start of every cycle. WALK also changes once per cycle, but steps up or down from its previous

value rather than leaping anywhere at all, so it wanders instead of jumping, changing direction when it reaches the top or the bottom of its range.

- **SYNC button and RATE knob:** these two work together. With Sync off, the Rate knob sets a frequency in hertz, from 0.02Hz (one cycle every fifty seconds) up to 20Hz (twenty cycles per second). Click the Sync button and the same knob instead chooses a rhythmic division, from 2/1 down to 1/32, including dotted and triplet values. The knob's readout shows the current setting either way.
- **PHASE knob:** shifts the point in the cycle at which the oscillator's output is read, displayed in degrees. Use it to offset one LFO against another running at the same rate, or against your host's transport when Sync is on. Double-click it to return to zero.
- **LEVEL knob:** how far the oscillator swings, displayed as a percentage. This is the depth control. At 100% it swings the destination half of its full range to either side of where the knob is sitting, which amounts to the whole range when the knob is centered; at 50% it swings half as far. Double-click it to return to 100%.
- **LAYER menu:** which layer the LFO affects, either one of the six or all of them. Unlike a MIDI connection, an LFO offers no "current" setting; LFOs are attached to their assigned layer or all of them at once.
- **DESTINATION menu:** the setting the LFO modulates. The list is shorter than the one a continuous MIDI connection offers, confined to the destinations that make sense under a repeating signal: SPEED, START POSITION, END POSITION, OUTPUT LEVEL, PANNING, BUSS SEND, DELAY SEND, REVERB SEND, FILTER FREQ, FILTER RES, and SLICE #. Leaving it at UNASSIGNED doesn't stop the oscillator but leaves it driving nothing.
- **OUT RANGE slider:** sets limits on the modulated value. The two thumbs are the lowest and the highest the destination will be driven to, whatever the oscillator and the control's own setting would between them otherwise produce. At their default positions, hard against the two ends of the slider, they impose no limit at all. Use them to keep a sweep within the range of values that are useful, e.g. to stop an LFO on Output Level from ever quite arriving at silence.

SLICE # is the exception to all of this. Slices are positions rather than a knob setting, so an LFO pointed at Slice # jumps the play head to a slice chosen outright rather than nudging it relative to anything, with the Level knob scaling the span of slices it covers and the Out Range thumbs confining it further.

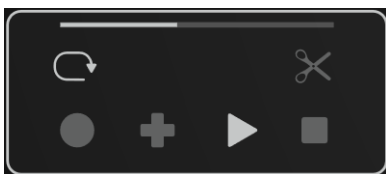
The display at the top right of the panel draws the shape as it runs, with a dot riding along it at the oscillator's actual rate, so you can see what the connection is doing at a glance. It follows the Level knob, so you can watch the swing shrink as you turn the level down. Random and Walk draw the values they really produced, as a staircase scrolling from right to left with the newest value at the right-hand edge. The closed rectangle in the Modulations view animates in the same way, so you can keep an eye on all of the LFOs at once without opening any of them.

A synced LFO takes its position from your host's transport whenever the transport is rolling, so it keeps its place in the bar and arrives at the same point in its cycle every time you loop a section. When the transport is stopped it free-runs at whatever rate its division works out to at the current tempo, rather than freezing until you press play again.

Finally, note that the LFOs drive Circa's controls directly and deliberately write nothing to your host's automation, for the same reason the Gesture Recorder doesn't: thirty oscillators sending a stream of parameter changes would fill your DAW's undo history with junk edits and write automation you never asked for.

Gesture Recorder

Circa's Gesture Recorder captures and plays back your interactions with Circa's controls. It operates independently of the audio recording. While there happen to be six slots in the gesture recorder and six audio layers, there's no direct connection between the Nth gesture slot and the Nth layer.



The gesture recorder appears on the Modulations view in the lower half of Circa's window. Each slot in the gesture recorder looks like the illustration here.

The four familiar symbols along the bottom of the slot's panel do what you'd expect: clicking them makes the gesture recorder record, overdub, play, and stop. Clicking the stop button a second time resets the gesture's playback position to its beginning. The oval-arrow

switch near the upper-left corner determines whether the gesture plays over and over or stops after playing once. Once you've recorded a gesture, you'll see a thin grey line at the top of the panel. When you play a gesture, this line is overlaid with a growing white line to indicate the gesture's current position within its recording.

To record a gesture, click the record button then move one or more controls in Circa's window. Everything you do will be recorded, in real time, until you click another of the slot's buttons. At that time the length of the recording will be fixed. If you click either the play or overdub button, the gesture will start playing from the beginning, reproducing all of your manipulations of the controls.

The overdub control lets you add new interactions to the gesture as it plays. Any new control change gets added to the existing gesture recording. If you engage the record button while a gesture is playing, any and all activity in the gesture is erased as it plays, and any new activity is recorded. Recording is isolated from playback: playing one gesture while recording another does not copy the playing gesture's moves into the new recording, and a gesture cannot re-record its own playback. What you actually touch is what gets captured.

When you record a gesture, there's usually some lag between when you hit the record button and when you move the first control. Like everything you record, this lag will be recorded as part of the gesture and may seem like a delay between when you play the gesture and when something happens. This is particularly noticeable if you're using a mouse-based computer and you must switch between Circa's views to reach the first control. Clicking the little scissors button near the top right of the gesture's slot trims off that interval at the beginning of the recording, so that the first control change happens immediately after you click the gesture's play button.

A long press on the stop button erases the gesture altogether.

Almost all of Circa's controls and parameters can be recorded in gestures. Most of the exceptions are there to avoid confusing outcomes: switching views in Circa's window, interactions with the modulation settings, and the gesture recorder's own controls cannot be recorded.

Gestures can be played and stopped with modulation mappings, so that you can use MIDI controllers to start and stop gestures. See the Modulations section above for more information.

Gesture playback moves Circa's own controls without notifying your host of every step, so replaying a gesture won't fill your DAW's undo history with edits you didn't make yourself or write unwanted automation into a record-armed track. When a gesture stops, your DAW is brought up to date with a single clean edit for each parameter the gesture touched, so automation lanes and generic parameter views show the true final values. If you happen to be dragging one of those controls at that instant, your hand wins. Controls you move by hand automate and undo normally, as always.

Because the gesture recorder's intention is to record your manual interactions with Circa, you may find that controlling some of Circa's parameters with the gesture recorder is not as smooth as using your host DAW's automation features. You may also find the opposite to be true. This depends on many things, including your particular host's approach to automation and your audio interface buffer size.

Effects

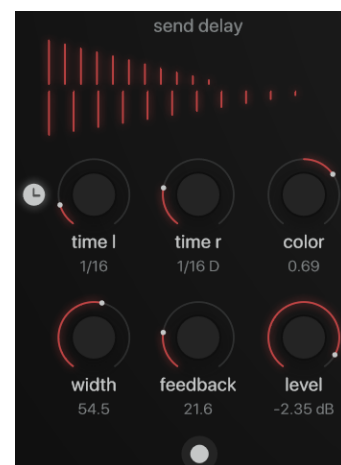
Circa includes four audio effects—delay, reverb, equalizer, and compressor—designed to complement its looping capabilities. These effects enhance your loops with creative textures, spaciousness, tonal shaping, and dynamic control. They're useful whether you're subtly polishing Circa's mix or dramatically transforming your sound.

Delay

Leftmost in Circa's Effects view is a stereo delay, with separate delay lines for each channel. Delays can be used for anything from short doubling effects to long, spacey echoes. The delay has the following controls:

The TIME L and TIME R knobs set the delay times of the left and right delays respectively. They have a range of one millisecond to two seconds. Turning on the SYNC switch—the small icon that looks like an analog clock—makes the delay times operate in metrical units, expressed as fractions of a measure. The letters D and T indicate dotted and triplet values, so for example a setting of 1/8 D means a dotted eighth note. When the Sync switch is on, the delay times change to match the tempo of your host DAW.

The FEEDBACK knob sends some or all of the delayed signals back to the inputs of the delay lines. If the feedback is zero, you'll hear only a single delayed copy of the signal. Turn up the Feedback and you'



ll hear more copies which fade out with each repeat. A setting of 100 means that all of the delayed signal goes back to the inputs, sustaining the sound more or less indefinitely.

A set of filters modifies the tone of the delayed signal. The COLOR knob controls these filters, brightening or darkening the sound of the delay. Turn it clockwise to reduce the low frequencies, making the sound thinner and brighter. Turn it anti-clockwise to reduce the highs, making the sound darker and warmer.

The two delay lines operate independently, potentially creating a wide stereo effect. Delays panned completely to the left and right sometimes sound too wide, so we've added a handy WIDTH knob. At 100% the delays are entirely separate, creating the widest separation. Turn the knob down to move the delays towards the center. A setting of zero gives a mono output.

The LEVEL knob adjusts the loudness of the delayed signal. Turn it all the way anti-clockwise and you won't hear the delay at all. This knob is also present on the Levels view for convenience.

Click the dot below the Feedback knob to turn the delay on and off altogether.

Reverb

Reverberation, or reverb for short, is the reflections we hear when a sound bounces off the walls, floor, ceiling, and other objects around us. Artificial reverb can add realism to an electronic sound or create unnatural effects. Circa's reverb originated with our experience with Adverb and Eos, our dedicated reverb plugins, but is updated and improved to work with a wide variety of sounds.

The DIFFUSION knob controls how much the initial density of echoes, often known as early reflections, builds up over time. Turning up the diffusion increases the build-up of echoes immediately following the original signal.

The DECAY knob controls how long it takes for the reverberated sound to fade out. This control has the greatest influence on the overall sound of the effect. Its range is 0.1 to 10 seconds, which is approximately the amount of time it takes the reverb generated by a full-volume signal to fade to silence.

The SIZE knob varies the apparent size of the simulated acoustic space. As will be evident when you listen to it, rotating the Size knob changes the bigness of the reverberated sound.

The COLOR knob is a simple-to-use tone control. Turning it to the left makes the reverb's sound darker; turning it to the right makes the sound brighter.

The LEVEL knob adjusts the loudness of the reverb. Turn it all the way anti-clockwise and you won't hear the reverb at all. This knob is also present on the Levels view for convenience.

Click the dot below the Decay knob to turn the reverb on and off altogether.



Main Buss Equalizer

Circa's equalizer is a straightforward but powerful three-band EQ, designed to quickly shape and enhance your sound. It provides independent control over the low, mid, and high frequencies. It affects all of the signals on the main stereo buss, including the delay and reverb.

The LOW and HIGH gain knobs adjust the volume of the bass and treble frequency bands respectively. Each knob allows boosting or cutting by up to 20 dB, giving plenty of control for subtle adjustments or bold tonal shaping. The frequency switches for the Low and High bands let you select different corner frequencies, adjusting the range where the bass and treble controls take effect.

The MID band is semi-parametric, offering even greater flexibility. The Mid gain knob provides 20 dB of boost or cut, while the FREQ knob lets you dial in precisely which midrange frequency you want to adjust. This allows you to quickly focus on key elements of your sound, from emphasizing vocal clarity to taming an unwanted resonance.

Click the dot below the Freq knob to bypass or engage the EQ.



Main Buss Compressor

Circa's compressor reduces the dynamic range of audio signals, making loud parts softer and quiet parts louder. Derived from our wildly popular Rough Rider compressor, it's capable of smoothing out performances, controlling peaks, and adding punch or sustain to Circa's mix. The compressor follows the EQ and is the last stage of Circa's signal path.

The THRESHOLD knob sets the level at which compression starts. When the input signal rises above this threshold, the compressor activates and begins reducing gain. Lowering the threshold makes the compressor act on quieter sounds, while raising it means only louder peaks will be compressed.

The RATIO knob determines how aggressively the compressor reduces signal levels once the threshold is crossed. For example, a 2:1 ratio means that signals exceeding the threshold by 2 dB will only increase the output by 1 dB. Higher ratios yield stronger compression, with ratios above 10:1 effectively acting as limiters.

The ATTACK knob controls how quickly the compressor responds when the signal exceeds the threshold. Shorter attack times clamp down immediately, taming transient spikes. Longer attack times let more of the initial punch through, preserving transient detail.

The RELEASE knob determines how fast or slow the compressor stops reducing gain once the signal falls below the threshold. Shorter release settings quickly restore the original volume, resulting in a more aggressive or pumping sound. Longer release settings produce smoother and more natural results.

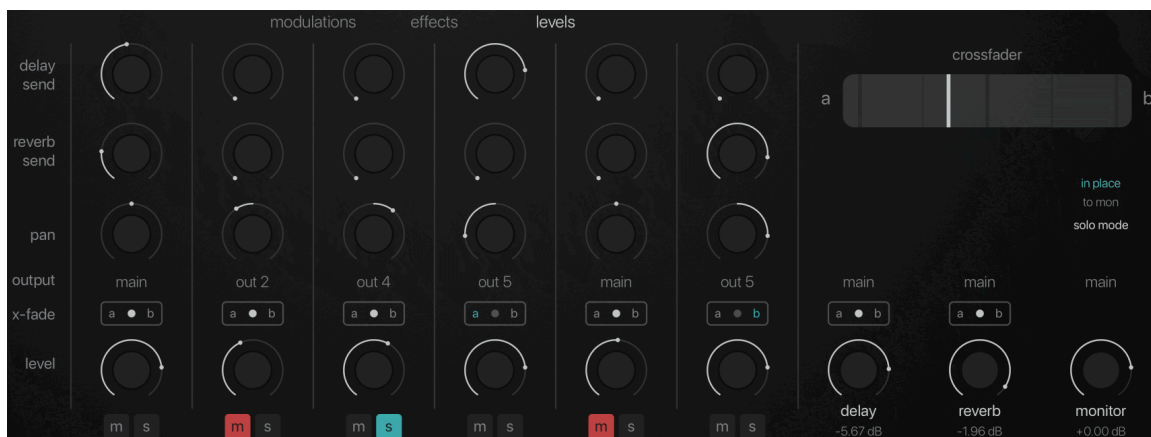
The MAKEUP knob compensates for any overall volume loss caused by compression. Turning this knob up boosts the compressed signal, allowing you to match or surpass the original loudness. This is useful for maintaining consistent levels when comparing compressed and uncompressed audio.

Click the dot below the Release knob to bypass or engage the compressor.



Levels

Circa's Levels view provides mixer-like control over each of the plugin's six independent looping layers. Each layer has its own dedicated channel strip, allowing precise management of audio routing, levels, and effects sends.



At the top of each strip are two knobs labeled DELAY SEND and REVERB SEND, which control how much of each layer's audio is sent to Circa's built-in delay and reverb. Turn these knobs clockwise to add space and depth to individual loops, or turn them fully anti-clockwise to keep the loops dry.

Below the send knobs, the PAN knob adjusts the stereo position of each loop within the stereo field. Turning this knob left or right places the loop audio accordingly, helping you create width and separation in your mix.

The OUTPUT selector determines the destination for each layer's audio output. Use it to assign loops to different outputs for additional processing or routing within your host DAW. There are six output busses, labeled MAIN and OUT 2 through OUT 6. Host DAWs vary in their handling of multiple-output plugins; some require specific settings before the additional outputs are accessible.

Each channel strip also has a LEVEL knob that adjusts the volume of the layer. Turn this knob clockwise to increase loudness or anti-clockwise to lower it.

Finally, each channel strip includes M and S buttons. Clicking the M button mutes the layer, removing it temporarily from the mix. Clicking S solos the layer. The SOLO MODE switch at the right of the pane determines what soloing does: with IN PLACE, solo works the way you'd expect, muting all of the other layers. With TO MON, the soloed layer is sent to the monitor output instead, without respecting mutes and without disturbing anything else—handy for auditioning a layer through headphones while the mix continues undisturbed.

The Crossfader

Beneath the channel strips is a CROSSFADER, marked A at its left end and B at its right. Each channel strip has an X-FADE selector with three positions: A, THROUGH, and B. The delay and the reverb have X-FADE selectors of their own.

Anything set to THROUGH—which is the default—ignores the crossfader entirely and passes to the output unaffected. Anything assigned to A or B is placed under the crossfader's control: move the crossfader toward A and the layers assigned to A come up while those assigned to B fade away, and vice versa. The blend follows an equal-power law, so a sweep from one side to the other holds a steady overall loudness rather than dipping in the middle.

This gives you a DJ-style way to trade between two groups of loops. Assign a few layers to A and a few to B, leave the rest on Through as a constant bed, and you can move between two arrangements with a single gesture—one that can be recorded into the gesture recorder or driven from a MIDI controller like any other parameter.

Effects and Monitor Levels

In the lower right corner of the Levels view you'll find knobs for adjusting the loudness of the delay and reverb. These knobs duplicate the level knobs in the Effects view; we added them to the Levels view to make it easy to adjust the loudness of the effects while adjusting the output settings of the layers. Likewise, the MONITOR knob duplicates the knob of the same name in the Master Input area of the Layers view.

There is an output selector above the delay and reverb level knobs. By default the effects are added to the main output buss, but you can route them to the five other output busses or the cue buss. The output selector above the Monitor knob chooses whether the monitor signal is added to the main buss or the cue buss.

Automation

Most of Circa's parameters can be automated using your host's automation features. Consult your host's documentation for information on how to use these features. Some of Circa's controls, most notably the Trigger button, cannot be automated. Momentary controls like this one do not work well with all hosts and plugin formats. A workaround is to use a modulation assignment to operate the Trigger with a MIDI message.

Recording Time and Memory Usage

Circa has no fixed limit on the length of its recordings. You can make loops as long as the available memory in your computer permits. Circa allocates extra memory as it records to provide its undo/redo mechanism. If Circa thinks it will run out of memory while recording it employs a couple of tactics to compensate. It uses the same tactics while loading presets.

The first time memory runs low, Circa disables its undo/redo mechanism and releases the extra memory consumed for this feature. A warning message appears at the top of the window and the Undo button disappears, but recording continues without interruption and you can continue to use all other features of Circa. If memory runs low again, more drastic measures are employed: recording immediately stops, no further recording is permitted, and the warning message changes accordingly. You can play existing recordings and even overdub, but you can't record new layers.

As a defensive mechanism, once Circa enters either of these low-memory fallback states, it will remain in this state until you load a preset or completely close Circa and open a new instance. Hence it is smart to immediately save your work, in either a new preset or via your DAW's session, if you see either of these warnings. Once you've done that you can continue to use Circa without fear of losing work. If you free up memory on your system by, for example, closing other applications, you may find that you can reload your preset or session without Circa entering one of its low-memory states.

Monome Grid and Arc Control

Circa has built-in support for the *grid* (128-button version) and *arc* hardware controllers made by monome³, providing a more hands-on experience with looping. You can control many of Circa's operations with the buttons and knobs, and the LEDs provide visual feedback. Setting up takes a few simple steps.

First, if you haven't done so already, install serialosc. Circa, and most other things, depends upon serialosc to communicate with monome hardware. The instructions and links to the relevant installers are available on the monome site, here: <https://monome.org/docs/serialosc/setup/>

Next, turn on the SEARCH FOR MONOME AND/OR ARC ON INSTANCE switch in Circa's Settings view. After turning it on, restart the plugin. Different hosts have different features for restarting plugins; many DAWs provide a power switch which you can toggle off and back on. If all else fails, delete Circa altogether and add a new instance.

Your operating system may ask for your permission to access network devices. This is because serialosc uses UDP networking commands to communicate via OSC. Circa doesn't talk to other devices on your network, but serialosc needs your permission to use networking commands when it starts up, so grant it this access if it asks.

Once you've done this, Circa will automatically connect to your grid and/or arc, and these switches will appear at the top of its window:



In this illustration, there is a monome grid connected but no arc connected. You can connect or disconnect Circa from your devices by clicking on the buttons on the left.

The mappings between Circa and the grid and arc are described in a separate document, available here: <https://github.com/Audio-Damage-Inc/manuals/blob/main/monome-arc-layout-guide.pdf>

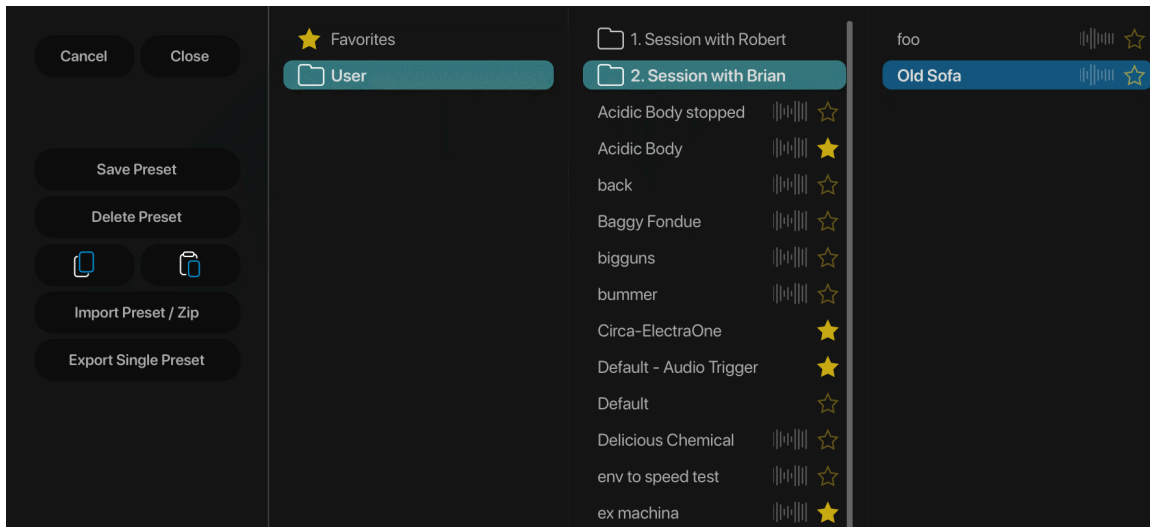
A few things to bear in mind. Circa's monome integration supports only one instance of Circa at a time. It works only in the desktop versions: there is no serialosc or libmonome on iOS, so while the code is present in the AUv3, it sits idle. Finally, Logic, GarageBand, and MainStage are heavily sandboxed and do not allow a plugin to see the local network, so Circa will not discover your monome hardware in those hosts. This applies to the AudioUnit, VST3, and CLAP versions alike—it's the host, not the plugin format, which imposes the restriction.

³See <https://monome.org> for information about monome and their products and creations.

Presets

Unlike most plugins, Circa does not come with a collection of presets, in rather the same manner that a turntable doesn't come with records. It's a blank canvas for your creations; of these, the ones you wish to keep become your presets.

Circa has a pane called the preset browser for saving and subsequently retrieving your presets. Click the floppy-disc icon at the top-right corner of the window to open it. Click anywhere outside of the browser to dismiss it.



The browser displays presets and folders in scrollable lists, arranged in columns. The leftmost list shows the categories within Circa's preset collection: FACTORY, which is empty in a new installation and reserved for any content we distribute later, and USER, which is where your own work goes. Once you have marked at least one preset as a favorite, as described below, a FAVORITES category appears above them.

Clicking any folder reveals its contents in the next list, and those folders can contain sub-folders of their own. The browser displays four columns, so you can nest up to three levels of folders within the Presets folder—that is, two levels of folders within the User folder. Clicking on a preset name loads the settings into Circa. Double-clicking a preset name loads the preset and dismisses the preset browser.

Loading a preset irretrievably erases Circa's current settings and any audio you've recorded, so if you have created something that you want to use again, save it as a new preset before loading another preset.

Click the CLOSE button to dismiss the browser. If you click CANCEL instead, the browser closes and Circa's settings revert to the state they were in when you opened it. Once you have clicked on any item in the panel you can navigate with the keys on your keyboard: the left and right arrow keys move the selection between columns, and the up and down arrow keys move it within a list. Tapping the ESC key has the same effect as clicking Cancel.

The stars to the right of the preset names mark presets as favorites. Click an empty star to mark a preset, and click a filled star to remove the mark. Once you've marked at least one favorite, the Favorites category appears in the leftmost column; clicking it shows you everything you have marked, gathered from wherever it happens to live, and clicking a name loads it as usual.

You can move presets and folders between folders by dragging and dropping them. Drag a preset onto a folder in another column to move it into that folder, or drop it onto an open column to move it into the folder that column represents.

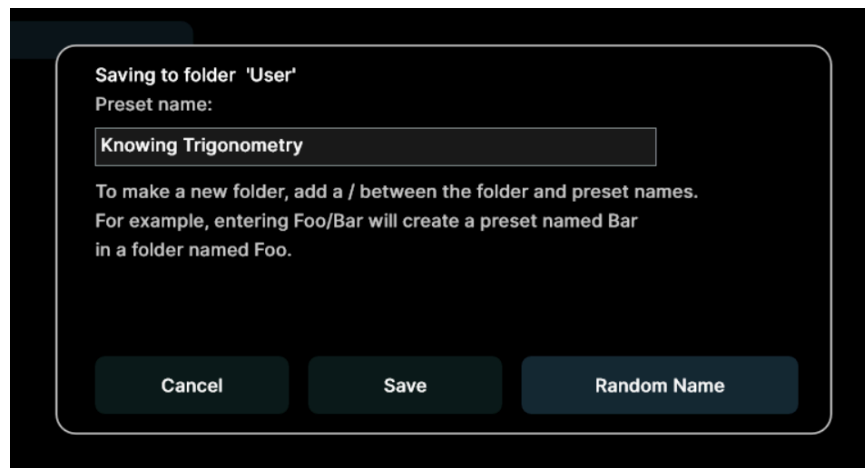
Where Presets Live

The folders and presets in the browser correspond to folders and files within Circa's own folder on your storage device. This folder is located at `C:\ProgramData\Audio Damage\Circa\` on Windows, and `~/Music/Audio Damage/Circa/` on macOS and Linux. You can store your presets anywhere you like, but for them to show up in the User list they must be placed in the User folder within Circa's folder. Also, to avoid possible collisions during future updates, do not store your presets within the Factory folder.

Any folders you create within the User folder show up as folders in the User list. You can create sub-folders within the User folder, and sub-folders within those, but you can't nest them any deeper than that because the preset browser has only four lists.

Saving Presets

To save your presets, click the SAVE PRESET button at the left edge of the window. This invokes a dialog box with a couple of helpful features. As the text therein describes, you can create a folder within the destination folder—whose name is given at the top of the dialog box—by adding the folder’s name to the beginning of the preset’s name, separated by a slash mark.



Clicking the RANDOM NAME button replaces the preset’s name with a pair of words chosen at random from two lists. While the resulting names won’t have any connection with what the plugin is doing, you may find this feature useful for coming up with alternatives to routine names like “My Preset 12”.

Potential pitfall: once you’ve saved a preset, clicking its name in the list loads the preset, overwriting whatever changes you’ve made since you saved it. Hence if you want to save the preset again to preserve your changes, **do not** click on its name before saving it.

You can delete presets and folders from the lists by clicking their name and then clicking the DELETE PRESET or DELETE FOLDER button. Circa will give you a chance to confirm this action or cancel it. If you confirm, the preset or folder will be removed from your storage system and is gone for good.

Importing and Exporting Presets

Preset files are plain-text XML files, so you can exchange them online in forums, copy them between a Windows computer and a Macintosh (and even between an iPad and a regular computer), email them to your friends, and so on.

The two buttons with icons representing copying and pasting—copy on the left, paste on the right—copy Circa’s current settings to the system clipboard and paste settings from the clipboard. You can use the copy and paste commands to transfer settings between two instances of Circa, or paste the settings into an email message or text editor. When copied to the clipboard, presets are presented in the same XML text as used in preset files.

The IMPORT PRESET / ZIP button provides a way to add presets to Circa without manually moving them into the appropriate folders in your file system. Clicking this button produces a file-browser window wherein you can select either a single preset file or a .zip file containing one or more presets. After you select the file, Circa copies the presets into whichever folder you have selected in Circa’s preset list, unzipping the file first if necessary.

Depending on whether you’ve selected a preset or a folder, the EXPORT SINGLE PRESET or EXPORT FOLDER AS ZIP button performs the complementary function. First select either a preset or a folder in Circa’s list, then click the export button. A file-save window appears; choose a location in your file system, give the file a name, and click Save. If you have chosen a folder in Circa’s preset list, the plugin places it and all of the presets it contains in a .zip file.

Default Preset

If you save a preset with the special name “Default” in the User folder, new instances of Circa will load it automatically when you add it to your DAW session. You can use a default preset file to give yourself the same starting point with Circa whenever you use it. Circa’s installer creates a default preset file for you, but feel free to replace it with your own.

Presets and Audio

Circa optionally stores the audio contents of its layers in presets and in your DAW sessions. The appropriately named switch in the Settings view, described previously, governs whether or not this happens.

Obviously it's nice, and customary, for presets and DAW sessions to restore all of the settings and internal states of a plugin. The drawback to storing audio within presets is that the file sizes can become quite large, as Circa stores audio in a lossless, uncompressed, 24-bit format. You may notice that it takes longer to load DAW sessions than usual if you have one or more instances of Circa containing a number of lengthy layers. The workaround for this is to record Circa's output onto regular audio tracks in your DAW, rather than depending upon Circa to store the audio.

If a preset contains audio, it is marked with a little waveform icon to the right of its name.

And Finally...

Thanks for purchasing Circa. We make every effort to ensure your satisfaction with our products and want you to be happy with your purchase. Please write to support@audiodamage.com if you have any questions or comments.

Document Revisions

- 7 April 2025: Initial release
- 30 April 2025: Various corrections
- 10 June 2025: Removed discussion of fixed recording length, added discussion of low-memory handling; changed MIDI CC values for momentary controllers (now 0-63 and 64-127 rather than 0 and 127); pointed out pitfall with inadvertently loading a saved preset
- 1 July 2025: Updates for software release 1.1. Added What's New section, added description of Gesture Recorder, added new targets in Modulations, updated screenshots where needed
- 14 August 2025: Added information about monome grid and arc integration
- 5 December 2025: Added note about lack of drag and drop in AUv3 on iOS
- 19 September 2026: Updates for software release 1.2. Added the LFOs, Hold Slice and the slice presets; rebuilt in our current manual format